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Mohamed H. Gad-Elrab

Senior AI Engineer with a focus on developing multi-agent and GenAI systems. Expertise in LLM applications, NLU, Neuro-Symbolic AI, and Knowledge Graphs. Proven track record in delivering impactful AI solutions, mentoring talent, and contributing to AI publications and IP portfolio. Dedicated to fostering innovation and creating production-ready systems with strong technical leadership, a dynamic strategy, and a hands-on approach.

Key Technical Competencies

- **Large Language Models (LLMs):** Prompt engineering, fine-tuning, and integration with enterprise systems.
- **AI Agents & GenAI Workflows:** Building multi-agent systems with LangGraph, AutoGen, and GraphRAG.
- **Knowledge Graphs & Semantic Technologies:** Ontology modeling, SPARQL, link prediction, graph-based reasoning.
- **Neuro-Symbolic AI:** Designing hybrid systems combining symbolic reasoning and deep learning.
- **Machine Learning:** Model design, training, and assessment using PyTorch, Scikit-learn, Transformers, PyKEEN ... etc.
- **API & System Development:** strong Python knowledge and libs for prototyping, such as FastAPI, Streamlit, Chainlit.
- **Automation Workflows:** Frameworks for data and execution automation utilizing Airflow and Argo.
- **DevOps:** Provisioning cloud infrastructure on Azure, AWS, using Docker, Kubernetes, Terraform, and CI/CD pipelines.
- **Vector & Hybrid Search:** Semantic and vector-based search on (un-)structured data using Elastic-search and Neo4j.
- **Software Engineering Processes:** including requirement gathering, Agile methodologies *e.g.* Scrum, and tools like Jira.

Professional Experience

Dec 2024 – present
Lead AI Research Engineer, Bosch Center for AI, Renningen, Germany
Semantic Understanding and Reasoning Research Group
- Leading the design & development of multi-agent AI systems to optimize internal enterprise workflows.
- Collaborating with business unit experts to analyze their needs and align them with GenAI technologies.
- Organizing a focus group for the R&D teams to build competencies in Agentic AI and GraphRAG.
- Supervising master's and PhD students, contributing to Bosch's publications and patent portfolio.

Jan 2021 – Nov 2024
Research Engineer, Bosch Center for AI, Renningen, Germany
Natural Language Processing and Neuro-Symbolic AI Group
- Successfully built a portfolio of PoCs, Demonstrations, and services for Neuro-Symbolic AI & NLU.
- Provisioned and managed cloud infrastructure & tooling to support Neuro-Symbolic AI and LLMs R&D.
- Led Bosch use-case development within the innovative EU Horizon project: "Graph Massivizer".

Apr 2015 – Dec 2020
Research Assistant, Max-Planck-Institut für Informatik, Saarbrücken, Germany
Databases and Information Systems Group (Prof. Gerhard Weikum)
- Research areas: information extraction, knowledge representation, and explainable AI.
- Tutored several post-graduate courses and co-supervised master thesis students.

Apr 2019 – Jul 2019
Research Intern, Bosch Center for AI, Renningen, Germany
Natural Language Processing and Semantic Reasoning Group
- Authored two research papers at tier-1 venues, resulting in three patent applications.

Jul 2012 – Sep 2013
Research Engineer, A-tech Group, Cairo, Egypt
- Researched methods for real-time indoor Augmented Reality experience for mobile devices

Education

2015–2020
PhD Candidate, Max-Planck-Institut für Informatik, Saarbrücken, Germany
- Thesis: "Explainable methods for knowledge graph refinement and exploration via symbolic reasoning"
  Supervision: Dr. Daria Stepanova & Prof. Gerhard Weikum.

2013–2015 **M.Sc. in Computer Science, Saarland University**, Germany
- Full scholarship from **Intl. Max-Planck Research School for Computer Science** (IMPRS-CS).
- Thesis: "AIDArabic+: Named-Entity Disambiguation for Arabic Text"
  Supervision: Dr. Mohamed Amir Youssef & Prof. Gerhard Weikum.
- Awarded **Günter-Hotz Medal** for the outstanding academic performance. GPA: 1.1/1 (A with Honors).

2007–2012 **B.Sc. in Computer Science and Engineering, German University in Cairo (GUC)**, *Egypt*
- **Full scholarship** awarded as a top-10 student in the Egyptian Secondary School Certification.
- Thesis: "Saliency-based Scene Text Detection"
  Supervision: Dr. Faisal Shafiat & Prof. Andreas Dengel (**DFKI, Kaiserslautern**, Germany).
- 3rd on Computer Science Dean's List, GPA: 0.83/0.7 (A+ with Highest Honors).

Other Activities

- **Scientific Reviewing**: Conferences: AAAI'18 '19, ESWC '17- , ISWC'18- . Journals: IEEE Access'19, IP&M'19, AIJ'20.
- **Tutorial Co-organizer:** "Neuro-Symbolic AI for Industry 4.0" (NeSyAI4) at ISWC'23
- **Recent Invited Talks:**
  - "Neuro-Symbolic AI & KGs Applications in Industrial Settings", Passu Uni., Germany.
  - "Applications of KG Link Prediction in Production and Business at Bosch", Nile Uni., Egypt.
  - "Graphs and RAG for Better Industrial GenAI Solutions" co-speaker, ICKG, Abu Dhabi, UAE.
- **Side Projects:** Deep Learning Model for Predicting Stock Movements, Self-hosted Smart Home.

Publications and Patents

Author of 25+ publications in leading data and semantics venues, including ISWC, CIKM, and WWW, along 10+ active patent applications. Below is a selection of key publications:

- **No-Code ML Pipeline Development: Leveraging Knowledge Graphs and Language Models.**
  Sarah Sajid, Antonis Klironomos, Evgeny Kharlamov, Frank Hopfgartner,Mohamed Gad-Elrab. *ESWC2025 Demos*.
- **Rule-based Knowledge Graph Completion with Canonical Models**
  Simon Ott, Patrick Betz, Daria Stepanova, Mohamed H Gad-Elrab, Christian Meilicke, Heiner Stuckenschmidt. *CIKM2023*.
- **Rule-based Knowledge Graph Completion with Canonical Models**
  Simon Ott, Patrick Betz, Daria Stepanova, Mohamed H Gad-Elrab, Christian Meilicke, Heiner Stuckenschmidt. *CIKM2023*.
- **A Study on Entity Linking Across Domains": " Which Data is Best for Fine-Tuning?**
  Hassan Soliman, Heike Adel, Mohamed H. Gad-Elrab, Dragan Milchevski, Jannik Strötgen *RepL4NLP@ACL2022*.
- **Utilizing Language Model Probes for Knowledge Graph Repair**
  Hiba Arnaout, Trung Tran, Daria Stepanova, Mohamed H. Gad-Elrab, Simon Razniewski, Gerhard Weikum. *Wiki2022*.
- **ExCut: Explainable Embedding-based Clustering for Knowledge Graphs**
  Mohamed H. Gad-Elrab, Trung-Kien Tran, Daria Stepanova, Heike Adel, Gerhard Weikum. *ISWC2020*.
- **ExFaKT: Framework for Explaining Facts using Knowledge Graphs and Text**
  Mohamed H. Gad-Elrab, Daria Stepanova, Jacopo Urbani, Gerhard Weikum. *WSDM2019*.
- **Rule Learning from Knowledge Graphs Guided by Embedding Models.**
  Vinh Thinh Ho, Daria Stepanova, Mohamed H. Gad-Elrab, Evgeny Kharlamov, Gerhard Weikum *ISWC2018*.
- **Named Entity Disambiguation for Resource-Poor Languages**
  Mohamed H. Gad-Elrab, Mohamed Amir Yosef, Gerhard Weikum. *ESAIR Workshop, CIKM 2015*.

Personal Information

Born November 12, 1990 in Asiut, Egypt
Nationality German
Languages Arabic (Native), English (Proficient user), German (Independent user)